

Balarakesh Gundapaneni

 Balarakesh G |  Balarakesh Gundapaneni |  balarakeshg16@gmail.com |  +91 8184930950

EDUCATION

2023 - 2027 Bachelor's Degree at **SRMAP University** (GPA: 9.56/10.0)
2021 - 2022 Class 12th Board of Intermediate Education, Andhra Pradesh (982/1000)

SKILLS

- **Languages, Web & Databases:** Python, JavaScript, C, C++, HTML, CSS, SQL, PostgreSQL, MongoDB
- **Frameworks & Libraries:** React.js, Node.js, Express.js, OpenCV, MediaPipe
- **AI/ML & NLP:** Machine Learning, Natural Language Processing (NLP), Transformers, Deep Learning
- **Developer Tools & Platforms:** Git, GitHub, Postman, Docker, Supabase, Vercel, Firebase

EXPERIENCE

- Intern at AddWise Tech Innovation** **May 2025 – July 2025**
- Engineered a **digital certification management suite** catering to **100+ learners across 10+ institutes**, featuring **OTP-based authentication**, Supabase integration, and dynamically generated QR-coded PDFs.
 - Deployed asynchronous mail workflows and **serverless certificate distribution** through Vercel functions, ensuring end-to-end delivery within **5 seconds** and full audit traceability.
- Intern at NighaTech Global** **August 2025 – April 2026**
- Architected **claverIt**, a unified personal-business ecosystem supporting **4 operational modes** with trust-priority caller identification and consent-gated contact intelligence.
 - Orchestrated **role-based interaction workflows** enabling user-business chat, appointment booking, and calendar-based reminders with event-driven notifications via **OneSignal**.

PROJECTS

- QubitChat** [GitHub](#)
- Architected an **AI-powered document intelligence system** for PDFs and images, generating embeddings via **Gemini API** and **ChromaDB** with **50+ semantic chunks per document**.
 - Devised a **hybrid retrieval pipeline** combining classical similarity search and **Grover's Algorithm (Qiskit simulation)** with automatic fallback for datasets up to 2^{10} **vectors**.
- Apadhbandhav** [GitHub](#)
- Engineered an **IoT-based accident detection and emergency response system** supporting **5 user roles** (normal user, police, hospital, admin, super admin) with automated crash event capture and alert initiation.
 - Orchestrated a **multi-stage emergency escalation workflow** triggering alerts via **SMS and voice calls**, expanding search radius from **5 km to 10 km, 20 km, and up to 40 km** until responders are found.
- IntruderVision** [GitHub](#)
- Developed an **AI-powered surveillance system** using **OpenCV** and **MediaPipe**, processing video streams through **2 concurrent threads** for motion tracking and intruder detection.
 - Automated evidence capture, **email alerting**, and cloud archival across **3 system stages** (detection, notification, storage) to support continuous monitoring.
- HandWave Unlock** [GitHub](#)
- Designed a **gesture-driven access control system** recognizing **3 personalized hand patterns** using **MediaPipe**-based hand tracking.
 - Implemented a **real-time gesture validation pipeline** comprising **3 processing stages** (landmark extraction, pattern matching, access decision) for secure unlock control.